J-space Phase 3 — final state of record

Output protection, content specificity, and bridge routing

Frozen design `jspace-phase3-freeze-v1` · release audit 2026-07-31

Scope. This handout is the release state, not the chronological lab notebook. Frozen outcomes were never rewritten. Corrections and sensitivities are event-sourced in `reports/evidence_events.jsonl`; exact paths and hashes are in `phase3_release_manifest.json`.

1 Executive result

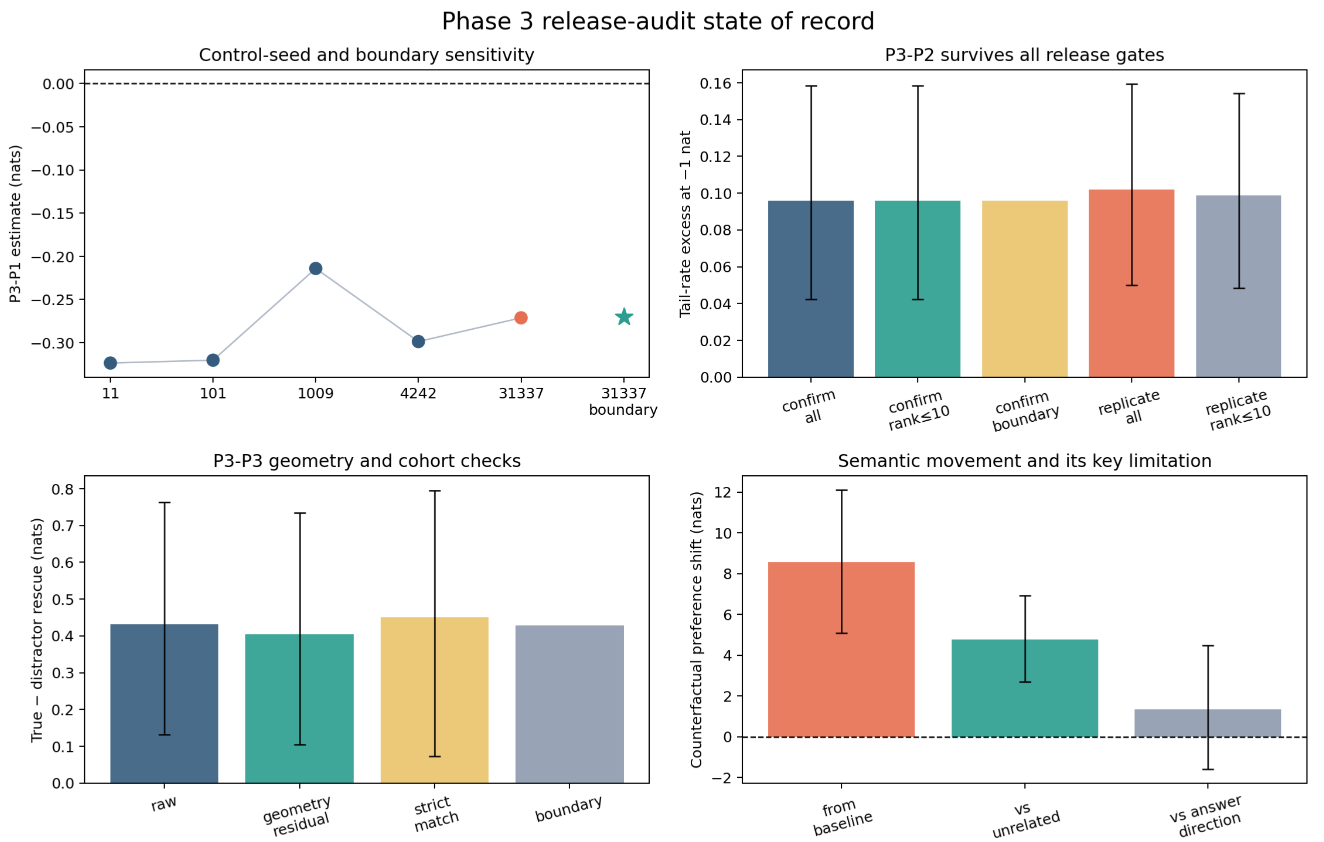
Phase 3's strongest result is narrow and reproducible: **on Qwen, span-safe J-space ablation produces a content-specific heavy tail beyond an exact rank-and-energy matched control**. It survives clean-answer protection, five Qwen control seeds, boundary-safe cohort grading, accepted-alias aggregation, and a held-out family replication.

result	confirmatory state of record	held-out replication	licensed status
P3-P1	−0.271183, exact $p = 0.057892$, 17 families	−0.197020, exact $p = 0.219482$, 17 families	descriptive; negative and seed-sensitive
P3-P2	+0.095833, $p = 1/100001$, 188 items / 26 families	+0.102083, $p = 1/100001$, 190 items / 28 families	**confirmatory and replicated**
P3-P3	+0.431367 nats, $p = 0.009180$, 94 items / 26 families	not collected	confirmatory chosen-distractor contrast; not replicated

The state reconstruction replaces Qwen's unrecorded historical `hash(item_id)` control realization with explicit `sha256-v1` seed 31337. OLMo rows remain the immutable historical controls. This leaves P3-P2 unchanged but makes P3-P1 explicitly descriptive.

Evidence: `p3-control-seed-contract-audit-v2`

Evidence: `p3-n8-p3-level3-qwen36-27b-v1`



2 What the release audit corrected

methods

1. **Protected-answer stratum.** The preregistered clean-rank filter was absent from the frozen analysis. Recomputed baseline-only ranks show the P3-P2 estimate is still $+0.095833$ for the exact scored alias (183/188 items) and any accepted alias (185/188), each $p = 1/100001$.
   Evidence: `p3-protocol-audit-protected-answer-qwen-v1`
2. **Control realization.** The historical Python hash salt is unrecoverable. Five full Qwen seeds leave P3-P2 identical but move P3-P1 across the .05 threshold; no "near-miss" inference is licensed.
3. **Inference objects.** P3-P1 now uses all $2^{17} = 131{,}072$ family sign patterns. At the historical control, its randomization-compatible 95% set is $[-0.535135, +0.014565]$. The percentile-t interval is $[-0.537109, +0.015201]$; the old estimate$\pm 1.96\,SE$ interval is labeled a normal approximation.
   Evidence: `p3-inference-audit-v1`
4. **Actual Phase 3 reproduction.** N8-P3-L1 reproduced 61/61 quantities exactly. Stratified L2 GPU cells pass on all three models. L3 reproduces all 188 Qwen rows, all deterministic arms, and the stable control exactly.
5. **Metadata and registry.** The held-out analysis tier was corrected event-wise to `phase3-replication`. Historical N8 jobs were relabeled N8-P2. One obsolete N8-P2 output whose filename was later overwritten is withdrawn; the actual N8-P3 artifacts are intact.

3 Boundary, cohort, and accepted aliases

methods

3.1 Generation boundary and cohort

Boundary-safe grading rejects four substring false positives and recovers four Unicode-accent matches among 2,916 G5 rows. A deterministic, model/family/alias-length-stratified review of 100 positives and 100 negatives has zero disagreements. At state seed 31337:

view	historical cohort	boundary-safe cohort
P3-P1	−0.271183	−0.270160
P3-P2	+0.095833 (188)	+0.095833 (186)
P3-P3	+0.431367 (94)	+0.428230 (93)

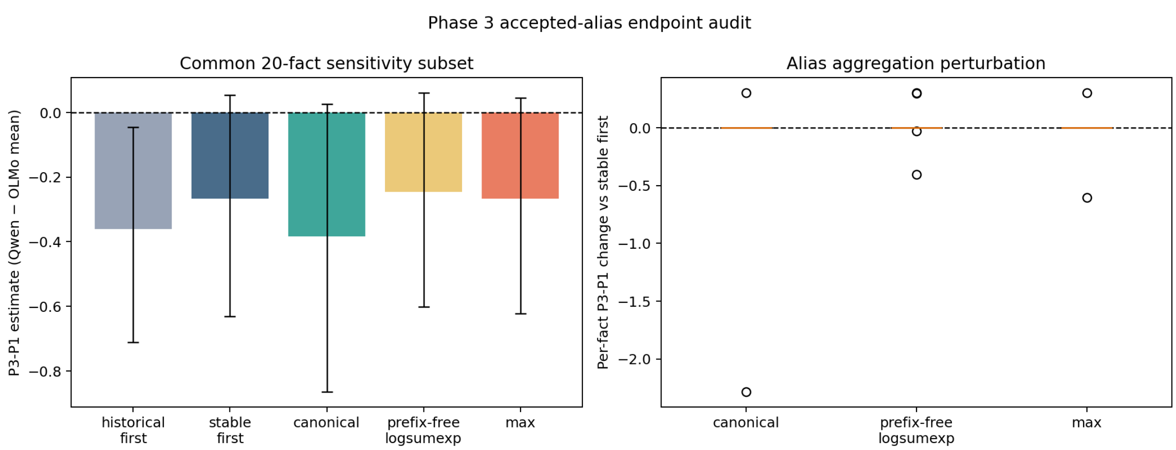
Superset population views report missing eligible outcomes rather than pretending they are identified.
Evidence: `p3-boundary-cohort-sensitivity-v2`

3.2 Accepted-alias endpoint

The alias subset was frozen before outcomes: 20 common facts, all 17 shared families, and all four multi-alias facts. Each model scored 52 alias cells on the RTX PRO 6000. Token-prefix collisions are zero and first-alias replay errors are below 0.000214 nats.

endpoint	P3-P1 subset	change vs stable first
stable first alias	−0.267301	reference
canonical alias	−0.383906	−0.116605
prefix-disjoint logsumexp	−0.246098	+0.021203
max alias (diagnostic)	−0.267357	−0.000056

Canonical sensitivity is confined to the two facts whose canonical form differs from alias zero. Qwen's P3-P2 subset tail is identical under all views. Phase 4 uses frozen prefix-disjoint logsumexp prospectively.
Evidence: `p3-alias-endpoint-cross-model-v1`



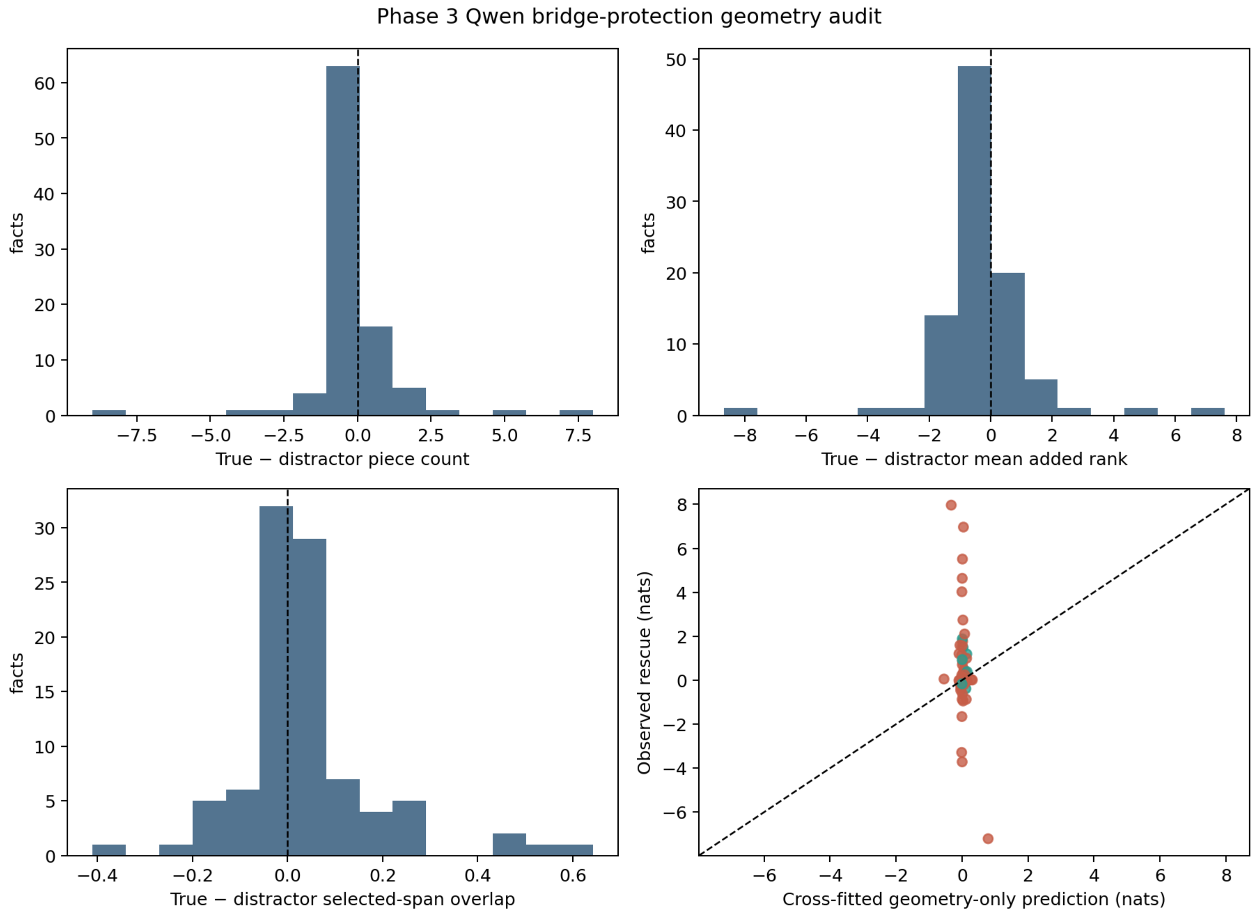
4 Bridge mechanism closeout

4.1 Chosen-distractor rescue

phase3-confirmatory / methods

The corrected geometry audit replays baseline, span-safe, true-bridge, and distractor-bridge arms exactly across 94 facts and 67,834 layer-position rows. Rank-accounting failures and final selected/protected overlap are zero.

Raw rescue is +0.431367 nats [+0.132018, +0.763437]. A nested leave-one-family-out geometry model has cross-fit $R^2 = -0.0947$; residual rescue is $+0.403816\,[+0.105071, +0.734497]$, $p = 0.01854$. The all-site exact geometry-matched subset has only 10 facts / 5 families (+0.450282, exact $p = 0.1875$) and is underpowered.
Evidence: `p3-bridge-geometry-qwen36-27b-v2`



The measured geometry does not explain away P3-P3. But the distractor was chosen semantically rather than randomized, and no untouched-family P3-P3 replication exists.

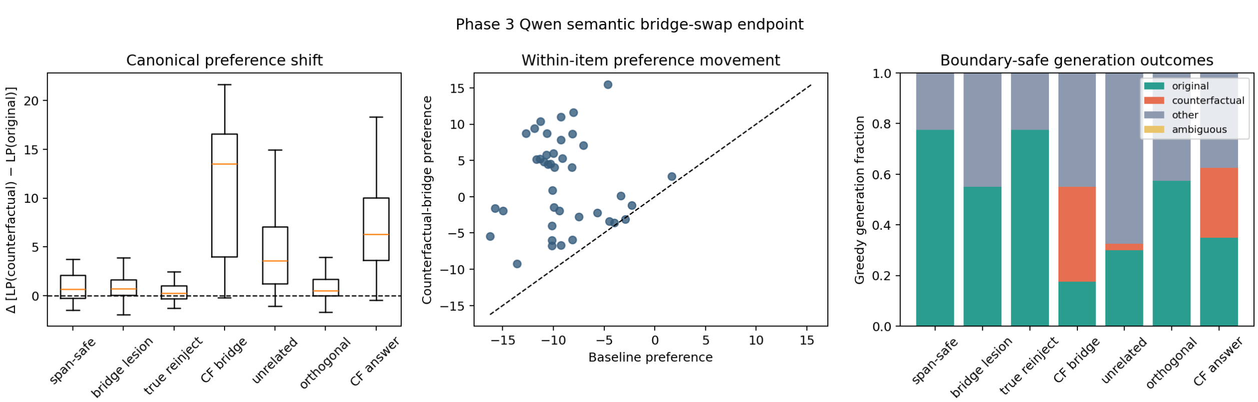
4.2 Counterfactual semantic movement

phase3-development

On the existing 40-fact / 13-family mediation cohort, counterfactual-bridge injection shifts $LP(\text{counterfactual}) - LP(\text{original})$ by $+8.582031$ nats $[+5.077661, +12.122991]$, exact $p = 0.000488$. Greedy output moves from 32 original / 0 counterfactual hits at baseline to 7 original / 15 counterfactual hits under the swap.

The key limitation is equally strong: direct counterfactual-answer direction injection shifts preference by $+7.240$ nats. Bridge minus answer-direction movement is only $+1.342254\,[-1.593275, +4.482051]$, $p = 0.419$. The result proves intended semantic movement but does not isolate an abstract bridge channel.

Evidence: `p3-bridge-swap-endpoint-qwen36-27b-v1`



5 Claim ladder and stopping boundary

1. **Confirmatory and replicated:** Qwen has a span-safe, content-specific J tail beyond matched dose.
2. **Confirmatory, not replicated:** true-bridge protection rescues more than frozen distractor protection; measured geometry does not account for it.
3. **Development:** counterfactual bridge injection moves answer probability and generation toward the intended target, but is not separable from answer-direction injection.
4. **Descriptive:** Qwen's direct-to-composed specificity shift is more negative than the OLMo pair's; P3-P1 is underpowered and control-seed-sensitive.
5. **Not licensed:** "selective global workspace" or broad working-memory claims. Prose damage exceeds task damage in standardized units, and Qwen's Bank-S composition term is null. The strongest noun is **knowledge-access channel**.

6 Limitations and next decisive experiment

- P3-P1 has 17 intersection families and unrecoverable historical control salts.
- P3-P3 lacks held-out-family replication and a randomized semantic distractor.
- The semantic endpoint is post-freeze development evidence on an existing cohort.
- Alias sensitivity has only four shared multi-alias facts.
- Capability filtering conditions on pre-intervention model behavior; missing superset outcomes remain unidentified.
- The intervention fails a behavioral selectivity guard on prose.

The next decisive experiment is a frozen untouched-family bridge bank that compares counterfactual bridge injection directly with a matched counterfactual-answer direction, alongside true, unrelated, and orthogonal controls. Phase 4 confirmatory work starts only after its own preregistration and freeze.

Binding Markdown: `reports/PHASE3_STATE_OF_RECORD.md`. Live evidence, raw parquet hashes, model/tokenizer/lens revisions, environment, seeds, and figure sources: `p3-release-manifest-v1`. Supersede, never edit.